A Theory of Economic Slack

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CHAPTER 1.

Introducing slack into business cycle research

In 1929, the Great Depression started in the United States. The event spurred a wave of research on recessions and business cycles. Beginning with Keynes, economists naturally concentrated on the depression's most devastating aspect: mass unemployment. A quarter of the labor force was thrown into unemployment, which had terrible consequences. In *The Grapes of Wrath*, Steinbeck (1939, pp. 317–321) describes:

> Carloads, caravans, homeless and hungry; twenty thousand and fifty thousand and a hundred thousand and two hundred thousand. They streamed over the mountains, hungry and restless—restless as ants, scurrying to find work to do—to lift, to push, to pull, to pick, to cut—anything, any burden to bear, for food. The kids are hungry. We got no place to live. . . . In the camps the word would come whispering, There's work at Shafter. And the cars would be loaded in the night, the highways crowded—a gold rush for work. At Shafter the people would pile up, five times too many to do the work.

Fifty years later, when the devastation of the depression had faded into memory, modern macroeconomics moved in a different direction: away from unemployment to focus on fluctuations in prices and quantities. But of course, unemployment surges remain a central feature of recessions. They return every once in a while, even if we had forgotten about them.

So this book returns the focus to economic slack—a concept that encompasses unemployment as well as other unsold and idle resources. As we go along we will see how different types of slack matter in modern economies.

The book starts by reviewing essential facts about slack and the business cycle. It then presents a theory to explain why slack emerges and why it fluctuates. Finally, it uses the theory to derive principles for the stabilization of slack fluctuations using monetary and fiscal policy.

1.1. Slack in Keynes's General Theory

Perhaps even more than other disciplines, macroeconomics is the product of its time. Trying to make sense of the Great Depression and offer advice to avoid such future calamities, Keynes (1936) wrote the *General Theory of Employment, Interest and Money*. This remains the most influential text in macroeconomics.

The *General Theory* offers a fairly balanced treatment of unemployment and inflation. It discusses inflation and unemployment to a significant extent, with slightly more coverage afforded to unemployment. In the book, inflation and deflation appear on 9 pages; unemployment is mentioned on 32 pages (table 1.1).

Although the *General Theory* does not cover forms of slack beyond unemployment, we know that slack was present in other forms during the Great Depression too: not only was it almost impossible for jobseekers to find jobs, but it was also almost impossible for shops to find customers and for factories to find buyers. As a result, many shops were empty, many factories were idle, and these businesses eventually had to lay their workers off and shut down. As Temin (2000, p. 301) wrote about the Great Depression: "unemployment rose to a peak of 25 percent and stayed above 15 percent for the rest of the 1930s;" at the same time, "there were many idle economic resources in America for a full decade."

This book presents a theoretical framework which shows us that these two phenomena are intrinsically linked. Slack on the product market—the fact that firms cannot find customers because aggregate demand is low—generates slack on the labor market—the fact that workers cannot find jobs because labor demand is low. A key insight of the book is that slack on the product market generates slack on the labor market. Plainly, if firms cannot sell their production, they will not hire workers.

In fact, it is not just that the phenomena of product market slack and labor market slack are linked. It is that they can be modeled entirely symmetrically. The same theory therefore applies to both types of slack, and both markets can be stacked to form a business cycle model with slack.

1.2. Absence of slack in New Keynesian texts

While Keynes was writing during a mass-slack event, by the 1970s, the economic environment had changed, leading to drastic changes in macroeconomic theory. Following the fast economic growth of the US economy in the 1950s and 1960s, and the spike in inflation

TABLE 1.1. Pages mentioning unemployment and inflation in the *General Theory* and New Keynesian texts, according to their indexes

	Keynes (1936)	Gali (2008)	Woodford (2003)
Unemployment	5–9, 13, 15, 16, 22, 109, 121, 128, 190, 235, 251, 274, 275, 278, 289, 334, 335, 346–349, 358, 359, 362–364, 381, 382	188	—
Inflation and related terms	119, 202, 207, 281, 291, 301, 303, 331, 332	1, 4, 11, 21–22, 31–32, 34, 37–38, 47, 49–52, 54–56, 60, 77, 79, 95–99, 103, 105, 107–108, 116, 120, 125–127, 130, 133–135, 137–139, 148, 155–156, 161–164, 168–176, 179, 189	3–5, 13–14, 17, 19–21, 39, 43–44, 46, 49, 90–101, 116–122, 126–138, 159–160, 176–177, 188, 203–205, 212–213, 215, 236, 248, 251, 262, 267, 272, 276–286, 289–295, 301–305, 317, 341, 347, 360–362, 381–382, 396–405, 408–409, 413–416, 418, 437, 440–441, 445–446, 460, 462–463, 467–485, 493, 499–501, 522–527, 544, 559–565, 576–582, 590–604, 615, 619–623, 639–641, 694–695, 714–715, 724–727
Number of unemployment pages	32	1	0
Number of inflation pages	9	60	195
Unemployment share	78%	2%	0%
Inflation share	22%	98%	100%

Besides the term "inflation," Keynes (1936) also features "deflation." In Gali (2008), the terms related to inflation include "inflation dynamics" and "inflation targeting." In Woodford (2003), the terms related to inflation include "inflation inertia," "inflation rates," "inflation-targeting central banks," "inflation-targeting rules," and "deflation." The unemployment and inflation shares are the shares of pages mentioning unemployment and inflation in the total number of pages mentioning either unemployment or inflation.

in the 1970s, the Old Keynesian model—which had formalized Keynes's ideas—was rapidly replaced by the Real Business Cycle model.

The Real Business Cycle model was built around Walrasian markets so it had no slack. Because it emphasized efficiency and monetary neutrality, it was not very useful for designing macroeconomic policies, and it was itself replaced by the New Keynesian model in the 1990s.

The New Keynesian model adopted the architecture of the Real Business Cycle model but swapped perfectly competitive markets for monopolistically competitive markets, which allowed researchers to introduce price rigidity and thus break efficiency and monetary neutrality. While the monopolistically competitive model produced rich price dynamics, it did not have room for unemployment or slack either, so the research squarely focused on inflation.

All in all, during the transition from Keynesian macroeconomics to New Keynesian macroeconomics, inflation was retained as a topic of interest but unemployment was lost. This omission of unemployment and slack and concentration on inflation appears clearly in New Keynesian textbooks. The leading New Keynesian texts are *Interest and Prices: Foundations of a Theory of Monetary Policy* by Woodford (2003) and *Monetary Policy, Inflation, and the Business Cycle* by Gali (2008). *Interest and Prices* discusses neither slack nor unemployment. *Monetary Policy, Inflation, and the Business Cycle* does not discuss slack, but unemployment is mentioned on one page at the end of the book, when possible extensions to the New Keynesian model are discussed.

Inflation, by contrast, is ubiquitous in both books (table 1.1). The term "inflation" appears on 97 pages in *Interest and Prices*. Additional related terms, such as "inflation inertia," "inflation-targeting central banks," and "inflation-targeting rules," appear on a further 98 pages. In total, there are 195 pages that cover inflation. In *Monetary Policy, Inflation, and the Business Cycle*, the term "inflation" appears on 52 pages. Related terms such as "inflation dynamics" and "inflation targeting" appear on 8 additional pages. In total 60 pages discuss inflation in that book.

1.3. Rebalancing business cycle theory

Policymakers in many countries have a dual legal mandate: keeping prices stable and maintaining the economy at full employment. Yet the New Keynesian model only speaks about price stability; it says nothing about full employment. The balanced treatment of unemployment and inflation that was a hallmark of Keynesian thought disappeared in the New Keynesian literature, as the focus turned solely to inflation.

The book offers a new business cycle model that provides a more balanced treatment of inflation and unemployment—one that can help us think both about price stability and full employment. The model accounts for fluctuations not only in quantities and prices, but also in slack. A unilateral focus on inflation is problematic in the context of business cycles because slack varies far more than prices over the business cycle, and it carries substantial welfare costs. So slack must be taken into account to provide an accurate depiction of business cycles and relevant policy advice.

So what, exactly, is economic slack? Economic slack describes productive resources in the economy that are unsold and therefore idle. There are many different forms of slack: people who cannot find a job and remain unemployed; machines left idle in a factory; employed workers left idle on the job; hotel rooms, restaurant tables, airplane seats that remain vacant; durable goods that cannot be sold and depreciate; nondurable goods that cannot be sold and perish.

And why is it crucial to consider slack when designing business cycle stabilization policies? The main reason is that fluctuations in slack have large consequences for welfare. Economic slack represents a waste of productive resources, and it is therefore something that should be limited. In addition to its wastefulness, unemployment generates other, large costs to society. People who are unemployed suffer from lower mental and physical health than employed workers. Even employed people in areas with high unemployment report lower well-being. Accordingly, good economic policy should avoid periods of elevated slack.

The next question is: How does the book build a business cycle model with slack? The central assumption in the model is that both labor and product markets are organized around a matching function. The matching function gives the number of trades achieved when a certain number of buyers and sellers look for each other (Petrongolo and Pissarides 2001). Because matching takes time and effort, not all sellers are able to trade, which naturally generates slack in both labor and product markets. This differs from canonical business cycle models: the Old Keynesian model features nonclearing Walrasian markets; the Real Business Cycle model features Walrasian markets; and the New Keynesian model features monopolistic markets.

The book's central argument is that understanding markets and business cycles requires embracing a framework built around a matching function. By using a matching function, the framework is realistic: it features slack on all markets. The framework is also tractable: markets can be studied using a supply-demand apparatus that incorporates buyers' and sellers' trading probabilities. Such a slackish framework not only explains the sources and costly nature of unemployment, but it also provides a more accurate lens through which to view markets, business cycles, and monetary and fiscal policy.

1.4. Overview of the book

Overall, the book is devoted to developing a new model of business cycles, and to exploring its implications for macroeconomic policy in the most transparent and practical way possible. The book is organized into five parts.

Part I finishes the introduction. Chapter 2 establishes the empirical foundation for the book, presenting essential but little-known facts about the existence, cyclicality, and social costs of slack. For readers interested in the sequence of research that led to the framework developed here, chapter 3 recounts that journey. It details the real-world puzzles and policy debates that motivated each step.

Part II develops the theoretical heart of the book: a model of a slackish market. Indeed, to understand why slack matters so much at the macro level, we must first understand how it operates at the level of individual markets. So, before bringing slack to business cycle analysis, the book starts at the market level: it develops a model of markets with supply, demand, prices, but also slack. We begin in chapter 4 with the key building block—the matching function. We then construct a basic slackish market model in chapter 5.

In the slackish market model, prices are set through norms. Because selling is difficult, sellers are glad when they can find a buyer. What this means is that there is a range of prices at which sellers would be willing to sell: the one they had in mind when they placed the good for sale, higher prices of course, but also lower prices that eat up some of their surplus. The same is true for buyers: it is not always easy to find exactly the right good, and buyers are happy when they can purchase what they were looking for, so they would be willing to pay a range of prices. Given these ranges, many price norms can arise in the model. At the same time, the price norm has critical implications for the behavior of the model. Therefore, in chapter 6, we examine how prices are set in the real world to isolate the main empirical properties of the price norm. Then, in chapter 7, we insert a realistic, somewhat rigid price norm in the slackish market model and explore its properties.

After that, we explore several extensions of the basic model. We explore how market capacity is determined (chapter 8), and how the model operates when buyers and sellers enter into long-term customer relationships once they have matched (chapter 9).

We then define market efficiency, which provides a welfare benchmark against which to measure market outcomes (chapter 10). The difficulty in matching buyers and sellers—illustrated by the presence of slack—implies that in reality we are very far from the auction market envisioned by Leon Walras. Many prices might prevail in markets: not just the efficient, market-clearing price that Walras focused on. The implication is that markets generally operate inefficiently. They are sometimes too slack, sometimes too tight, and efficient only in a knife-edge case.

Next, part III applies the slackish market model to the labor market. This is an important application because from a social perspective, unemployment is the most costly form of slack. There are also good data on labor market slack, which allows us to calibrate the model and compare its quantitative behavior with real-world patterns. In chapter 11, we build a slackish labor market model and show that it offers a unified theory of unemployment, including both job rationing and matching frictions. In chapter 12, we study the impact of labor-makret policies in the slackish labor market model: public employment, migration, and unemployment insurance. Last, in chapter 13, we compute the efficient unemployment rate in the United States over the past century, and the unemployment gap, to assess how far from social efficiency—and therefore full employment—the US labor market has been. We find that the US labor market has been generally inefficiently slack, and especially in recessions.

Having established that the unemployment gap observed in the US labor market is generally positive and sharply countercyclical, we must explain which macroeconomic forces generate such fluctuations and how macroeconomic policies—especially monetary policy—might stabilize them. This is what parts IV and V attempt to do.

Part IV scales up the model from a slackish market to a slackish economy. We start with a basic one-market economy in chapter 14 to understand the core properties of slackish business cycles. With this basic model, we see that aggregate demand shocks can generate countercyclical fluctuations in the unemployment gap, and that monetary and fiscal policy can counteract such fluctuations. In chapter 15, we introduce directed search into the business cycle model to produce a Phillips curve that links inflation to slack. In chapter 16, we extend the model to a more realistic economy with both labor and product markets. The two-market economy allows us to understand how slack percolates from the product market to the labor market.

With a complete business cycle model in hand, we turn in part V to the policy questions posed by the unemployment-gap evidence: How should monetary and fiscal policy be conducted when the economy is not at full employment? We derive formulas for the optimal conduct of monetary policy (chapter 17) and fiscal policy (chapter 18) based on the unemployment gap. The part closes by showing how slack data can be used to detect recessions early and therefore trigger an early policy response (chapter 19).

The book therefore addresses a number of questions that recur during all business cycles and that policymakers systematically face, especially in downturns. It explains for instance how much interest rates should fall when unemployment rises, how large stimulus spending should be in downturns, and how early policymakers should start worrying about an upcoming recession.

Part VI concludes the book. We begin by summarizing the main arguments and implications of the slackish framework (chapter 20). We then outline avenues for future research, highlighting how the concept of slack opens up new opportunities for data collection, modeling, and policy design (chapter 21). We also discuss how the models and policy insights developed in the book could be fruitfully applied to countries besides the United States—particularly developing countries, where slack is widespread. We close by situating the slackish business cycle model developed in the book within the macroeconomic landscape—drawing connections to Old Keynesian, Real Business Cycle, and New Keynesian models (chapter 22).

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